

## Comparing approaches for Enterprise Architects

There are at least 21 different approaches to modeling a complex evolving object, such as a business. These range from brain storming to more structured and formalized approaches.

The following table provides a quick reference guide to the approaches an enterprise architect can use to assist business operatives describe the business under the 3 sub phases of information architecture:

| Method     | Information architecture          |              |                         |                      |                   |                             |                |                              |            |        |
|------------|-----------------------------------|--------------|-------------------------|----------------------|-------------------|-----------------------------|----------------|------------------------------|------------|--------|
|            | Conceptual architecture           |              |                         |                      |                   | Logical architecture        |                | Physical architecture        |            |        |
| Bachman    | *                                 |              |                         |                      |                   | Entities & associations     | *DFDs          | Application generator        |            |        |
| *Bal sc    | Objectives                        |              | Strategies              |                      |                   | Data                        |                | *                            |            |        |
| *BPR 1     | Vision and Objectives             |              |                         | Baselines            |                   | *                           |                | Prototype                    |            |        |
| *BPR 2     | Ontology                          | Information  |                         | Function             |                   | Data                        |                | *                            |            |        |
| *BPR 3     | Plan                              |              |                         | Analyse              |                   | Design                      |                | Implement                    | Evaluate   |        |
| *BPR 4     | Plan                              |              |                         | Analyse              |                   | *                           |                | Plan                         | Evaluate   |        |
| *BPR 5     | Business modeling                 |              |                         | System design        |                   |                             |                | Implementation               |            |        |
| *Br storm  | 7 rules                           |              |                         |                      |                   | *                           |                | *                            |            |        |
| CoPR       | Transcendental doctrine of method |              |                         |                      |                   | Transcendental Logic        |                | Transc. doctrine of elements |            |        |
| FEAF       | Enterprise architecture           |              |                         |                      |                   | Segment architecture        |                | Solution architecture        |            |        |
| Geram      | Identification                    |              | Concept                 | Requirement          |                   | Design                      | Implementation | Build                        | Operate    | Change |
| I E        | Plan                              |              | Analyse                 |                      |                   | Design                      |                | Construct                    |            |        |
| *KM        | *Story telling - sharing ideas    |              |                         |                      |                   | Expert systems              |                | Knowledge repositories       |            |        |
| Macroscope | Strategy                          |              | Enterprise architecture |                      |                   | Enterprise value management |                | Busn transformation & change |            |        |
| NIST EA    | Business                          |              | Information             |                      |                   | Data                        | Applications   | Technical infrastructure     |            |        |
| PEAF       | Foundation                        |              | Management              |                      |                   | MetaModel                   | Governance     | Communication                |            |        |
| Ripose     | Strategic planning architecture   |              |                         |                      |                   | Logical architecture        |                | Solutions architecture       |            |        |
|            | Grammatical architecture          |              |                         | Systems architecture | Data architecture | Applications architecture   | Prototype      | Production                   |            |        |
|            | Objectives                        |              | Knowledge               |                      |                   |                             |                |                              |            |        |
| TOGAF      | *F&p                              | *Arch vision | *Busn arch              | *O&s                 | *Acm              | Information systems         |                | *Tech arch                   | *Migr plan | *I g   |
| T&Q        | Trivium                           |              |                         |                      |                   | Quadrivium                  |                |                              |            |        |
|            | Rhetoric                          |              | Grammar                 |                      |                   | Logic                       |                |                              |            |        |
| UML        | Foundation                        |              |                         |                      |                   | Behaviour                   |                |                              |            |        |
|            | Core                              |              | Auxiliary               |                      |                   | Data types                  |                |                              |            |        |
| Zachman    | Contextual                        |              | Conceptual              |                      |                   | Logical                     |                | *Phys                        | *A b       | *F e   |

How does one make up their mind as to which approach to use? Here are a few benefits you should look for in relation to the approach, namely the approaches:

- |                                                                                      |                    |
|--------------------------------------------------------------------------------------|--------------------|
| ▪ Effectiveness - capability; economic viability; unique features                    | - Common wealth    |
| ▪ Efficiency - speed of deliver; how practical it is to apply; how streamlined it is | - Common wellbeing |
| ▪ Ethics - equitability; honesty; transparency                                       | - Common good      |
| ▪ Ease of use - fluency; simplicity; intuitive                                       | - Common sense     |

### \*Notes:

- If we have missed an approach and you would like us to analyse it for comparison's sake, please email us at [info@ripose.com](mailto:info@ripose.com)
- Each phase may have multiple sub phases. Examine the method in detail for more information. We have fact sheets on most of these
- The grey area suggests no deliverable or phase could be identified
- Hyperlinks may or may not work - some approaches may have been removed
- Bal sc - Balanced scorecard
- There are about 5 different BPR (business process re-engineering) approaches
  - 1) Davenport & Short
  - 2) KBSI - Knowledge Based Systems, Inc
  - 3) ProSci - A BPR education series
  - 4) ECOPI - Electronic College of process innovation
  - 5) Proforma
- Br storm - Brain storming
- DFD - data flow diagram
- KM - Knowledge management
- TOGAF architectures - F&p = Framework & principles; Arch vision = Architecture vision; Busn arch = Business architecture
- O&s = Opportunities & solutions; Acm = Architecture change management; Tech arch = Technology architecture;
- Migr plan = Migration planning; I g = Implementation governance
- Zachman scopes - Phys = Physical; A b = As built; F e = Functioning enterprise

## A history of methodologies, frameworks and techniques

The following table shows a time line outlining the development of the approach:

Table 1: Developer by era

| Era           | Charlemagne | I Kant | Bachman | Codd | Dijkstra | MA Jackson | Drucker | Yourdon | Richter | Porter | Finklestein | Martin | Zachman | Fujitsu    | BPR | Geram | Kaplan | NIST EA | TOGAF | UML |
|---------------|-------------|--------|---------|------|----------|------------|---------|---------|---------|--------|-------------|--------|---------|------------|-----|-------|--------|---------|-------|-----|
| 782           | C de V      |        |         |      |          |            |         |         |         |        |             |        |         |            |     |       |        |         |       |     |
| 1755          |             | CoPR   |         |      |          |            |         |         |         |        |             |        |         |            |     |       |        |         |       |     |
| 1960s & 1970s |             |        | RDM     | 3NF  | SP       | JSD        | BSP     | SADT    | JSD 3NF |        |             |        |         |            |     |       |        |         |       |     |
| 1980s         |             |        |         |      |          |            |         |         | IA IE   | BSP    | IE          | IE     | ZF      | P+         |     |       |        |         |       |     |
| 1990s         |             |        |         |      |          |            |         |         | Ripose  |        | ZF          |        |         | Macroscope | BPR | Geram | BS     | Nist EA | TOGAF | UML |

Table 2: Developer, method and description

| Developer     | Method     | Description                                                   | Era         |
|---------------|------------|---------------------------------------------------------------|-------------|
| Bachman C     | RDM        | Role data model                                               | Early 1970s |
| BPR           | BPR        | Business process re-engineering                               | 1990s       |
| Charlemagne   | C de V     | Capitulare de villis - Governance of the royal estates        | 782         |
| Codd E        | 3NF        | Third normal form - normalisation                             | Late 1960s  |
| Dijkstra E    | SP         | Structured programming                                        | Early 1970s |
| Drucker P     | BSP        | Business strategic planning                                   | Early 1970s |
| Finklestein C | IE         | Information Engineering                                       | Early 1980s |
| Fujitsu       | Macroscope | Based on DMR's S+ P+ A+ B+                                    | 1987        |
| Geram         | Geram      | Generalised Enterprise Reference Architecture and Methodology | 1990        |
| Jackson M A   | JSD        | Jackson system development                                    | 1974        |
| Kant I        | CoPR       | Critique of pure reason                                       | 1755        |
| Kaplan R      | BS         | Balanced scorecard                                            | 1992        |
| Martin J      | IE         | Information Engineering                                       | Early 1980s |
| NIST          | NIST EA    | National Institute of Standards and Technology                | 1990        |
| Porter M      | BSP        | Business strategic planning                                   | 1980        |
| Richter C     | 3NF        | Learnt how to normalise                                       | 1975        |
|               | JSD        | Learnt how to structure a program from data                   | 1977        |
|               | SADT       | Studied SADT                                                  | 1978        |
|               | IE         | Information Engineering                                       | 1982        |
|               | IA         | Information architecture                                      | 1989        |
|               | Ripose     | Ripose                                                        | 1990        |
| TOGAF         | TOGAF      | The open group architecture framework                         | 1995        |
| UML           | UML        | Unified modelling language                                    | 1994        |
| Yourdon E     | SADT       | Structured Analysis and Design Technique                      | 1975        |
| Zachman J     | ZF         | Zachman framework                                             | 1982        |

## The information architect

The following table shows the sub classifications of an information architect and the skills an information architect needs to be a virtuoso in:

| Information architect                         |                        |                      |                   |                       |                          |                                  |            |                   |                      |
|-----------------------------------------------|------------------------|----------------------|-------------------|-----------------------|--------------------------|----------------------------------|------------|-------------------|----------------------|
| Conceptual /<br>enterprise/business architect |                        |                      | Logical architect |                       |                          | Physical /solutions architecture |            |                   |                      |
|                                               |                        |                      | Data architect    |                       | Application<br>architect | Data<br>base<br>admin            | Programmer | Test architect    |                      |
| Business<br>analyst                           | Knowledge<br>architect | Systems<br>architect | Data<br>modeler   | Data base<br>designer |                          |                                  |            | Systems<br>tester | Deployment<br>tester |

The stakeholders, skills, inputs, processes and outputs are as follows:

| Stakeholder     | Skill                 | Input                       | Process                           | Output                           |                                      |
|-----------------|-----------------------|-----------------------------|-----------------------------------|----------------------------------|--------------------------------------|
| Strategic mgt   | Business analyst      | Existing organisation chart | Refinement                        | Refined organisation chart       |                                      |
|                 |                       | Generic business statements | Facilitated sessions              | Business objectives              |                                      |
| Tactical mgt    | Knowledge architect   | Business objectives         | Facilitated sessions              | Knowledge model                  |                                      |
| Strategic mgt   | Systems architect     | Knowledge model             | Facilitated sessions              | Prioritised systems              |                                      |
| None            | Enterprise architect  | Refined organisation chart  | Prepare document                  | Business plan - proof of concept |                                      |
|                 |                       | Business objectives         |                                   |                                  |                                      |
|                 |                       | Knowledge model             |                                   |                                  |                                      |
|                 |                       | Prioritised systems         |                                   |                                  |                                      |
| Operational mgt | Data modeler          | Prioritised systems         | Facilitated sessions              | Logical data model               |                                      |
|                 |                       | Knowledge model             |                                   |                                  |                                      |
| None            | Data base designer    | Logical data model          | Rationalisation                   | Logical data base design         |                                      |
|                 |                       |                             |                                   | Subject area design              |                                      |
| Operational mgt | Application architect | Subject area design         | Rapid application design sessions | Logical applications             |                                      |
| None            | Data base admin       | Logical data base design    | Data base generation              | Physical data base               |                                      |
|                 | Programmer            | Physical data base          | Program code                      | Unit tested code                 |                                      |
|                 |                       | Logical applications        |                                   |                                  |                                      |
| Operational mgt | Systems tester        | Unit tested code            | Systems testing                   | Error free code                  |                                      |
|                 |                       |                             |                                   | Operating instructions           |                                      |
|                 | Deployment tester     | Error free code             | Stress testing                    | Production systems               |                                      |
|                 |                       |                             |                                   |                                  | Target hardware & software platforms |
|                 |                       |                             |                                   |                                  | Operating instructions               |
|                 |                       |                             |                                   | Update operating instructions    |                                      |

### Output content:

|                        |                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| Business objectives -  | Purpose statement; Benefits; Values; Performance indicators                                                   |
| Business plan -        | Financial budgets, risk analysis, production plan, quality assurance, governance, resource plan, project plan |
| Logical applications - | Screen designs, menus, reports, pseudo code                                                                   |